

KOMPLEXE ZAHLEN

Rechnen in $\mathbb{C}$. Berechnen Sie folgende Zahlen, sowie deren Konjugierte und Inverse:

* 1. $\frac{5}{3i-4} + \frac{13+i}{5(2-i)}$

3. " $\sqrt{i}$ " (also $z \in \mathbb{C}$ mit $z^2 = i$)

* 2. $\left(\frac{2-i}{1+i2}\right)^{30}$

4. $\left|\frac{74+i50}{1-i\sqrt{3}}\right|$

Lösung.

1. Nenner bekomme ich weg, indem ich mit dem Komplex-Konjugierten erweitere!

$$\frac{5 \cdot (-3i-4)}{(3i-4)(-3i-4)} + \frac{(13+i) \cdot \cancel{8} \cdot (2+i)}{5(2-i) \cdot \cancel{8} \cdot (2+i)} = \frac{5}{3^2+4^2} (-3i-4) + \frac{1}{5 \cdot (2^2+1^2)} (13+i)(2+i)$$

$$= \frac{1}{5} (-3i-4) + \frac{1}{25} (26 + 13i + 2i + \underbrace{i^2}_{-1}) = \frac{1}{5} \cdot \left[\underbrace{(-3i-4)}_{-1} + \underbrace{(5+3i)}_1 \right] = \frac{1}{5}$$

$25 + 15i = 5(5+3i)$

$$\frac{1}{5} = \frac{1}{5} \quad \left(\frac{1}{5} + 0 \cdot i = \frac{1}{5} - 0 \cdot i \right), \quad \left(\frac{1}{5} \right)^{-1} = 5.$$

2. $\left(\frac{(2-i)(1-2i)}{(1+2i)(1-2i)}\right)^{30} = \left(\frac{1}{1^2+2^2} (2-i)(1-2i)\right)^{30} = \left(\frac{1}{5} \cdot \underbrace{(2-4i-i+2i^2)}_{-5i}\right)^{30} = (-i)^{30} = \underbrace{((-i)^2)^{15}}_{-1} = -1$

$-1 = -1, \quad (-1)^{-1} = -1.$

Bei beliebigen komplexen Zahlen eignet sich die Polarform besser zum Multiplizieren bzw.

Potenzieren: $(r e^{i\varphi})^{30} = r^{30} e^{i(30 \cdot \varphi)}$

3. " $\sqrt{i}$ " = z : Suche z mit $z^2 = i$. Sei $z = x+iy$.

$$z^2 = (x+iy)^2 = x^2 + 2 \times ixy + (iy)^2 = x^2 + i2xy - y^2 = \underbrace{(x^2-y^2)}_{=0} + i \cdot \underbrace{2xy}_{=1} \stackrel{!}{=} i = \underline{0} + i \cdot \underline{1}$$

$$\Leftrightarrow \begin{cases} x^2 - y^2 = 0 \\ 2xy = 1 \end{cases} \Leftrightarrow \begin{cases} x^2 = y^2 \\ xy = \frac{1}{2} \end{cases} \Leftrightarrow \begin{cases} x = \pm y \text{ (I)} \\ xy = \frac{1}{2} \text{ (II)} \end{cases}$$

(I) in (II): $\pm y \cdot y = \frac{1}{2} \Rightarrow \pm y^2 = \frac{1}{2} \Rightarrow \underbrace{y^2}_{\geq 0} = \pm \frac{1}{2} \Rightarrow y = \pm \sqrt{\frac{1}{2}} \stackrel{!}{\Rightarrow} x = \pm y = \pm \left(\pm \sqrt{\frac{1}{2}} \right) = \pm \sqrt{\frac{1}{2}}.$

also zwei Lösungen $z = x + iy = \begin{cases} \sqrt{\frac{1}{2}} + i\sqrt{\frac{1}{2}} = z_1 \\ -\sqrt{\frac{1}{2}} - i\sqrt{\frac{1}{2}} = z_2 \end{cases}$

$$\bar{z}_1 = \sqrt{\frac{1}{2}} - i\sqrt{\frac{1}{2}} \quad z_1^{-1} = \frac{1}{\sqrt{\frac{1}{2}} + i\sqrt{\frac{1}{2}}} \cdot \frac{\sqrt{\frac{1}{2}} - i\sqrt{\frac{1}{2}}}{\sqrt{\frac{1}{2}} - i\sqrt{\frac{1}{2}}} = \frac{1}{\underbrace{\sqrt{\frac{1}{2}} + \sqrt{\frac{1}{2}}}_1} \cdot (\sqrt{\frac{1}{2}} - i\sqrt{\frac{1}{2}}) = \sqrt{\frac{1}{2}} - i\sqrt{\frac{1}{2}} = \bar{z}_1$$

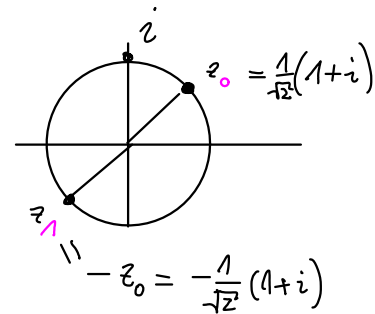
$$\bar{z}_2 = -\sqrt{\frac{1}{2}} + i\sqrt{\frac{1}{2}} \quad \text{Analog } z_2^{-1} = \bar{z}_2$$

später über Polarform:

" $\sqrt{i}$ " = z : Suche z mit $z^2 = i = 1 \cdot e^{i\frac{\pi}{2}}$

$$z_0 = \sqrt[2]{1} e^{i(\frac{\pi/2}{2} + 0 \cdot \frac{2\pi}{2})} = e^{i\frac{\pi}{4}}$$

$$z_1 = \sqrt[2]{1} e^{i(\frac{\pi/2}{2} + 1 \cdot \frac{2\pi}{2})} = e^{i(\frac{\pi}{4} + \pi)} = e^{i\frac{\pi}{4}} \cdot \underbrace{e^{i\pi}}_{-1} = -e^{i\frac{\pi}{4}}$$



$$\bar{z}_0 = e^{-i\frac{\pi}{4}} = z_0^{-1} \quad (\text{Probe: } z_0 \cdot \bar{z}_0 = e^{i\frac{\pi}{4}} \cdot e^{-i\frac{\pi}{4}} = e^0 = 1, \text{ d.h. } \bar{z}_0^{-1} = \bar{z}_0)$$

$$\bar{z}_1 = -e^{-i\frac{\pi}{4}} = z_1^{-1} \quad (\text{Probe: } z_1 \cdot \bar{z}_1 = (-e^{i\frac{\pi}{4}}) \cdot (-e^{-i\frac{\pi}{4}}) = e^0 = 1, \text{ d.h. } \bar{z}_1^{-1} = \bar{z}_1)$$

$$u. \left| \frac{74 + i50}{1 - i\sqrt{3}} \right| = \frac{|74 + i50|}{|1 - i\sqrt{3}|} = \frac{\sqrt{74^2 + 50^2}}{\sqrt{1^2 + (\sqrt{3})^2}} = \frac{\sqrt{5476 + 2500}}{\sqrt{4}} = \sqrt{\frac{7976}{4}} = \sqrt{1994} \approx 44,6...$$

$$\sqrt{1994} = \sqrt{1994}, \quad \sqrt{1994}^{-1} = \frac{1}{\sqrt{1994}}$$

OPER:

$$\left| \frac{74 + i50}{1 - i\sqrt{3}} \right| = \left| \frac{74 + i50}{1 - i\sqrt{3}} \cdot \frac{1 + i\sqrt{3}}{1 + i\sqrt{3}} \right| = \left| \frac{74 + i \cdot 74\sqrt{3} + i \cdot 50 + \overset{-1}{i^2} \cdot 50\sqrt{3}}{1^2 + \sqrt{3}^2} \right|$$

$$= \left| \frac{1}{4} \cdot (74 - 50\sqrt{3} + i \cdot (74\sqrt{3} + 50)) \right| = \sqrt{\left(\frac{1}{4} (74 - 50\sqrt{3}) \right)^2 + \left(\frac{1}{4} (74\sqrt{3} + 50) \right)^2}$$

$$= \sqrt{\frac{1}{16} \cdot (74^2 - 2 \cdot 74 \cdot 50\sqrt{3} + 50^2 \cdot \sqrt{3}^2) + \frac{1}{16} (74^2 \sqrt{3}^2 + 2 \cdot 74\sqrt{3} \cdot 50 + 50^2)}$$

$$= \sqrt{\frac{1}{16} \cdot (74^2 + 50^2 \cdot \sqrt{3}^2 + 74^2 \sqrt{3}^2 + 50^2)} = \sqrt{\frac{1}{4} (74^2 + 50^2)} = \sqrt{1994}$$

Eigener Lösungsversuch.

$$r e^{i\varphi}$$

Berechnen Sie $r = |z|$ und $\varphi = \text{Winkel zur } x\text{-Achse}$

Polarform. Schreiben Sie in Polarform:

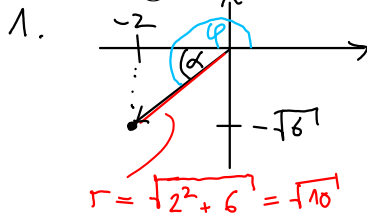
* 1. $-2 - i\sqrt{6}$

3. $-10i$

* 2. 10

4. $1 + (\sqrt{2} - 1)i$

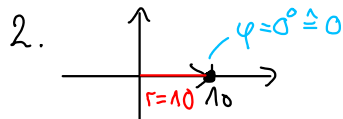
Lösung.



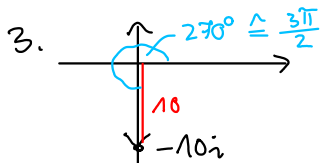
$$\tan \alpha = \frac{\sqrt{6}}{2} \Rightarrow \alpha = \arctan\left(\frac{\sqrt{6}}{2}\right) \approx 50,768^\circ \hat{=} 0,282 \dots \pi$$

$$\varphi = 180^\circ + \alpha = 230,768^\circ \hat{=} \pi + \alpha = \underline{1,282 \dots \pi}$$

$$\Rightarrow -2 - i\sqrt{6} = \underline{\sqrt{10} e^{i 1,282 \dots \pi}}$$



$$10 = \underline{10 \cdot e^{i \cdot 0} = 10 \cdot e^{i \cdot 2\pi} = 10}$$



$$\underline{-10i = 10 e^{i \frac{3\pi}{2}}}$$

4.

$r = \sqrt{(\sqrt{2}-1)^2 + 1^2} = \sqrt{2-2\sqrt{2}+1+1} = \sqrt{4-2\sqrt{2}}$
 $\varphi = \arctan\left(\frac{\sqrt{2}-1}{1}\right) = 23,5^\circ \hat{=} \frac{\pi}{8}$

$$\Rightarrow \underline{1 + (\sqrt{2}-1)i = \sqrt{4-2\sqrt{2}} e^{i \frac{\pi}{8}}}$$

Eigener Lösungsversuch.

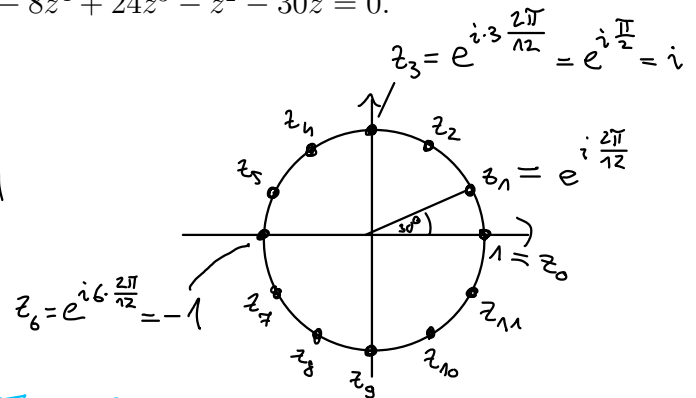
Komplexe Wurzeln und Nullstellen von Polynomen.

1. Was sind die 12. Einheitswurzeln?
2. Bestimmen Sie alle Lösungen von $z^5 + 4 = 4i$.
3. Bestimmen Sie alle Lösungen von $z^6 - 2z^5 - 8z^4 + 24z^3 - z^2 - 30z = 0$.

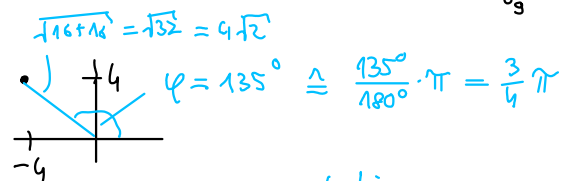
Lösung.

1. " $\sqrt[12]{1}$ ", d.h. Lösungen von $z^{12} = 1$

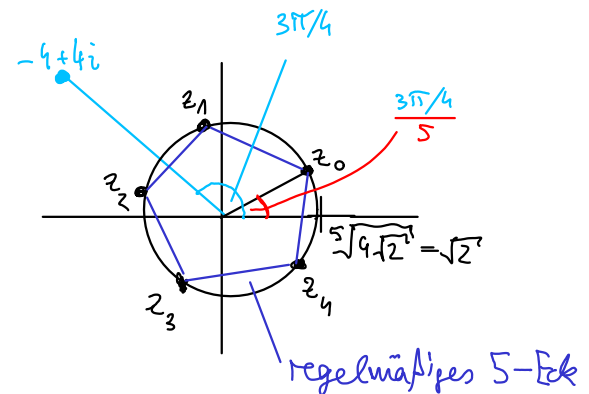
$$z_k = e^{i \cdot k \cdot \frac{2\pi}{12}}, \quad k = 0, \dots, 11$$



2. $z^5 = -4 + 4i = 4\sqrt{2} e^{i \frac{3\pi}{4}}$
 d.h. 5-te Wurzeln von $-4 + 4i$



$$z_k = \sqrt[5]{4\sqrt{2}} e^{i \left(\frac{3\pi/4}{5} + k \frac{2\pi}{5} \right)}, \quad k = 0, \dots, 4$$



3. Nach dem Fundamentalsatz der Algebra besitzt ein Polynom vom Grad 6 auch 6 NST, bloß welche?

$$z^6 - 2z^5 - 8z^4 + 24z^3 - z^2 - 30z = z \cdot (z^5 - 2z^4 - 8z^3 + 24z^2 - z - 30) \stackrel{!}{=} 0$$

$$\Rightarrow \underline{z_0 = 0} \quad \vee \quad z^5 - 2z^4 - 8z^3 + 24z^2 - z - 30 = 0$$

Raten mit Teilern von -30: Wertetabelle von -30 bis 30 in TR: $\underline{z_1 = -1}, \underline{z_2 = 2}, \underline{z_3 = -3}$

$$\text{Polynomdivision: } (z^5 - 2z^4 - 8z^3 + 24z^2 - z - 30) : (z+1) \cdot (z-2) \cdot (z+3) = \underline{z^2 - 4z + 5} \quad \text{s.u.}$$

$$\text{Davon NST } \underline{z_{4,5}} = \frac{4 \pm \sqrt{16 - 4 \cdot 5}}{2} = \frac{4 \pm \sqrt{-4}}{2} = \frac{4 \pm 2i}{2} = \underline{2 \pm i}$$

Eigener Lösungsversuch.

$$\begin{aligned}
 & (z^5 - 2z^4 - 8z^3 + 24z^2 - z - 30) : \underbrace{(z+1) \cdot (z-2) \cdot (z+3)}_{(z^2 - z - 2)} = \\
 & (z^5 - 2z^4 - 8z^3 + 24z^2 - z - 30) : (\underline{z^3} + 2z^2 - 5z - 6) = \underline{z^2} - \underline{4z} + \underline{5} \\
 & - (\underline{z^5} + 2z^4 - 5z^3 - 6z^2) \\
 & \quad \underline{-4z^4 - 3z^3 + 30z^2 - z} \\
 & \quad - (\underline{-4z^4} - 8z^3 + 20z^2 + 24z) \\
 & \quad \quad \underline{5z^3 + 10z^2 - 25z - 30} \\
 & \quad \quad - (\underline{5z^3} + 10z^2 - 25z - 30) \\
 & \quad \quad \quad \underline{ }
 \end{aligned}$$